

Hereditary Cancer Risk 15-Gene NGS Panel

Germline Sequence and Deletion/Duplication Analysis (Synthetic Demonstration)

Patient Demographics

Name: Brenda S. Sample
DOB: 09/30/1985 (Age 40)
Sex: Female
Patient ID: SYN-MRN-9912048

Ordering Clinician

Provider: Dr. Ethan Ross, MD
Facility: Pennsylvania Genetics Institute
NPI: 9847392015
Location: Philadelphia, PA

Specimen Chain of Custody

Specimen ID: SYN-SPEC-NGS-2098
Specimen Type: Whole Blood
Tube: EDTA Lavender Top (4.0 mL)
Collection Date: Jul 15, 2026
Received Date: Jul 16, 2026
Report Date: Jul 24, 2026

Result: Pathogenic Variant Identified

Status: Final
Gene: BRCA1 (Heterozygous)

Test Result Summary: A heterozygous pathogenic frameshift variant was identified in the BRCA1 gene in this synthetic sample. No other pathogenic alterations or copy number variants were detected in the remaining 14 genes tested.

Detailed Genetic Variant Findings

Gene Tested	LOINC	Classification & HGVS Variant	Zygosity	Interpretation
BRCA1 Genetic Analysis	69548-6	Pathogenic - c.5266dupC (p.Gln1756Profs*74)	Heterozygous	Abnormal (Pathogenic)
BRCA2 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
PALB2 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
TP53 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
CHEK2 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
CDH1 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
ATM Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal
MLH1 Genetic Analysis	69548-6	Negative / No Pathogenic Variant	N/A	Normal

Clinical Interpretation & Recommended Next Steps

The BRCA1 c.5266dupC (p.Gln1756Profs*74) variant is a well-characterized founder mutation in the BRCA1 tumor suppressor gene that results in premature protein truncation, confirming Hereditary Breast and Ovarian Cancer (HBOC) syndrome. Recommended next steps include:

- High-risk breast surveillance:** Annual contrast-enhanced breast MRI starting at age 25–30, alternating with annual mammography.
- Ovarian cancer risk reduction:** Consultation regarding risk-reducing bilateral salpingo-oophorectomy (RRSO) between ages 35–40.
- Cascade genetic testing:** Inform and offer targeted variant testing to all at-risk first-degree relatives.

Methodology & Limitations

Methodology: Genomic DNA is extracted from whole blood. Target enrichment of all coding exons and +/- 20bp flanking intronic boundaries of 15 hereditary cancer predisposition genes is performed using custom hybridization capture probes, followed by paired-end sequencing (Illumina NovaSeq). Copy number variations (CNVs) are called using normalized read-depth algorithms.

Limitations: This assay does not detect deep intronic mutations outside the targeted regions, complex chromosomal rearrangements, or low-level somatic mosaicism.

Laboratory Director Authorization: Dr. Sarah Jenkins, MD, FCAP | Electronically Signed | Date: Jul 24, 2026 | CLIA ID: 00D7654321